# Class NP

Exercise sheet 8

**Exercise 8.1.** Prove that the following problems are in NP:

(a) $\{\langle G, s, t, k\rangle \mid \text{graph } G \text{ contains a simple path from } s \text{ to } t \text{ visiting at least } k \text{ vertices}\}$;

(b) SUBSET-SUM $= \left\{\langle a_1, \ldots, a_m, t\rangle \in \mathbb{N}^{m+1} \text{ in binary} \mid \exists I \subseteq \{1, \ldots, m\} : \sum_{i \in I} a_i = t\right\}$;

(c) 3COLOUR $= \{G \mid \text{graph } G \text{ has a proper colouring using at most three colours}\}$.

**Exercise 8.2.** Suppose $L_1, L_2 \in$ NP. Is it true that $L_1 \cup L_2 \in$ NP? Is it true that $L_1 \cap L_2 \in$ NP?

**Exercise 8.3.** * Suppose that P = NP. Prove that a polynomial-time algorithm can find a clique of size at least $k$ in a given graph $G$, if one exists, and report that none exists otherwise.

**Exercise 8.4.** Recall that $L^* = \{w_1 \cdots w_r \mid r \geq 0,\ w_i \in L\}$. Suppose $L \in$ NP. Is it true that $L^* \in$ NP?